

ElectroMentor AI

Electrical Skills Assistant for TVET Trainees

Wiring/Circuit Guides · Safety Checklists · Photo Review · Task Tracker · AI Troubleshooting & Fault Detection

Version

1.0 — Draft

Date

August 2026

Platform

Web + Mobile App

Sector

Vocational Education (TVET) -
Electrical & Electronics

Status

Awaiting Approval

ElectroMentor AI is an AI-powered web and mobile platform purpose-built for polytechnic students, vocational trainees, apprentices, and electrical workshop learners in Bangladesh's TVET system. It gives trainees a wiring/circuit guide library, a safety checklist library, photo upload for review, and a practice task tracker, all powered by an AI Assistant for troubleshooting guidance, AI Vision for fault detection from photos, and AI-suggested safety checklists — bringing expert-level guidance to the workshop floor.

3 CORE AI FEATURES

AI Assistant <i>Troubleshooting guidance</i>	AI Vision <i>Fault detection from photos</i>	AI Suggestion <i>Safety checklist generator</i>
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DATA FLOW — INPUTS · AI ENGINE · OUTPUTS

Data Inputs • Photos of wiring/circuits • Trainee questions	AI Engine • AI Assistant troubleshooting model • AI Vision fault-detection (computer vision) • AI Suggestion safety-checklist engine	Outputs • Step-by-step guidance • Fault diagnosis • Safety checklist
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CORE USER FLOW



Product Goals • Give trainees on-demand access to wiring and circuit guidance • Detect wiring/circuit faults automatically from photos • Reduce reliance on instructor availability for troubleshooting • Reinforce safety practices with an always-available checklist • Track trainee practice tasks and skill progression over time	Target Users • Polytechnic Students — learn wiring/circuit fundamentals • Vocational Trainees & Apprentices — practice and troubleshoot tasks • Electrical Workshop Learners — review safety and technique
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FUNCTIONAL REQUIREMENTS

Feature	Description	Priority
FR1 — Wiring/Circuit Guide Library (Main Feature)	Trainee browses a library of wiring and circuit guides relevant to their coursework.	Must Have
FR1a — AI Assistant Troubleshooting (AI Feature)	AI Assistant answers trainee questions and gives step-by-step troubleshooting guidance in the context of a guide.	Must Have
FR2 — Photo Upload for Review (Main Feature)	Trainee uploads a photo of a wiring or circuit setup for review.	Must Have
FR2a — AI Vision Fault Detection (AI Feature)	AI Vision analyses the uploaded photo to detect wiring or circuit faults and returns a diagnosis.	Must Have
FR3 — Safety Checklist Library (Main Feature)	Trainee accesses a library of safety checklists relevant to electrical workshop tasks.	Must Have
FR3a — AI Suggestion Safety Checklist (AI Feature)	AI Suggestion generates a task-specific safety checklist based on the wiring/circuit work being performed.	Must Have
FR4 — Practice Task Tracker (Main Feature)	Trainee tracks completed practice tasks and reviews progress across the training curriculum.	Must Have

Release Timeline (MVP) - Month 1 — UX design, AI Vision model selection, guide/checklist data pipeline - Month 2–3 — Wiring/circuit guide library, AI Assistant troubleshooting - Month 4 — Photo upload, AI Vision fault detection, safety checklists - Month 5 — Pilot with 3 TVET institutes → public launch	Success Metrics (Month 6) - AI Vision fault-detection accuracy > 90% - Troubleshooting guidance resolution rate > 85% - Practice task completion rate > 60% - Weekly active trainee usage > 60% · NPS > 65
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